

GOURAB MODAK

+919073420524 gourabmodak28092003@gmail.com LinkedIn Github Codolio Leetcode Codechef Kolkata, India

INTERNSHIPS

AI/ML Core Team Member & DSA Mentor | Internship

Aug 2023 - Jun 2026
On-site • Jalpaiguri, India

Google Developer Student Clubs (GDSC)

- Mentored **100+** students in **Data Structures** and **Algorithms**, enhancing coding proficiency through bootcamps and workshops.
- Directed introductory **AI/ML** sessions and hackathons, boosting student participation in technical club events by **30%**.
- Collaborated with a team of **10** to organize tech fests, managing logistics and content delivery for **500+** attendees.

Technologies / Skills Used : Data Structures and Algorithms, Problem Solving, Machine Learning, Mentorship, Contest Discussions

Software Engineering Intern | Internship

May 2024 - Jul 2024
On-site • Mandi, India

iHub & HCl Foundation

- Engineered** an automated EEG signal processing pipeline for 64-channel ANT Neuro devices using **Python**, reducing manual noise-cleaning time by **40%** for 150+ patients.
- Spearheaded** development and benchmarking of 7 ML classifiers (SVM, RF, KNN), achieving **80%+** accuracy in detecting Parkinson's disease severity.
- Streamlined** model performance by **20%** through Grid Search hyperparameter tuning and integrated a QR-based lab attendance system to optimize workflows.

Technologies / Skills Used : Python, Machine Learning, Signal Processing, Statistics, Probability, EEG, EMG, Brain Computer Interface

PROJECTS

Pictionary — Real-Time Multiplayer Game | Github

Dec 2025 - Feb 2026

Individual Project

- Constructed a **real-time multiplayer system** with **Socket.IO**, hosting **50+** concurrent players across multiple rooms.
- Handled **100+** real-time socket events/second with sub-second synchronization and atomic state replication using **Redis**.
- Orchestrated round lifecycle management with **Redis-backed persistence**, ensuring **100%** state recovery during client reconnects.
- Devised session-based recovery and drawing history snapshots, restoring dropped users within **2 seconds**.

Technologies / Tools Used : TypeScript, Node.js, Socket.IO, Redis, Docker, MongoDB, Express, Tailwind CSS

AegisQueue — Distributed Job Queue System | Github

Dec 2024 - Jan 2026

Individual Project

- Architected** a distributed background job processing system supporting **concurrent multi-worker execution** with atomic MongoDB locking, ensuring **zero duplicate execution**.
- Validated** system throughput under concurrent load, sustaining **200+ jobs/minute** across multiple workers while preserving job-state integrity.
- Formulated** a state-driven lifecycle engine (QUEUED → RUNNING → SUCCESS) with retry enforcement, achieving **0% permanent job loss**.
- Instituted** heartbeat-based worker monitoring for stalled job recovery within **<30 seconds**, significantly enhancing fault tolerance.

Technologies / Tools Used : FastAPI, Python, MongoDB, Redis, Tailwind CSS, React, Node.js, Express

Competitive Programming Editor | Github Vercel

Aug 2025 - Sep 2025

Individual Project

- Developed** a browser-based IDE supporting **3+ languages** (C++, Python, Java and more) with **snippet templates** for competitive programming.
- Integrated** Monaco Editor (VS Code engine) to enhance **IntelliSense**, syntax highlighting, and auto-completion, boosting coding efficiency.

Technologies / Tools Used : React.js, Node.js, Tailwind CSS, Chakra UI, Monaco Editor

SKILLS

Databases :	MongoDB, PostgreSQL
Frameworks & Libraries :	React.js, Node.js, Express.js
Programming Languages :	C++, Python, Java, JavaScript, TypeScript, SQL
Tools & Platforms :	Git, GitHub, VS Code, Postman, Vercel, Docker

EDUCATIONS

Jalpaiguri Government Engineering College | Jalpaiguri, India

Jun 2022 - Jun 2026
CGPA : 9.33 / 10

B.Tech in Computer Science & Engineering

Indian Institute of Technology Madras | Chennai, India

Jun 2023 - Jun 2024
CGPA : 9.63 / 10

BS in Data Science (Foundational Level)

AWARDS & ACHIEVEMENTS

- Achieved** Global Rank 211 in TCS CodeVita 2025; **3-Star Coder** on CodeChef (Top 10%); solved **1500+** DSA problems.
- Secured** AIR 3967 in GATE 2025 (CS) among **170,000+** candidates (Top 2.3% Nationwide).
- Secured** Rank 2152 in WBJEE (Top 1%) and **94.2%**ile in JEE Main (Top 2% in Mathematics) in Engineering Entrances 2022.
- NPTel Silver Medalist:** Top 5% in Data Structures (Python), Joy of Computing, Discrete Mathematics.

RESEARCH PAPER PUBLICATIONS

AutoFedRL:A Dynamic Framework for Hematological Image Classification through Privacy-Preserving Collaborative Learning

Accepted
2025

International Conference on Data Electronics and Computing (ICDEC-2025) -- Springer

Towards Resource-Efficient Automation: A Comprehensive Survey on Sustainable IoT–Robotics Systems

Accepted
2025

National Conference on Advanced Computing in Sustainable Development (ACSD-2025)